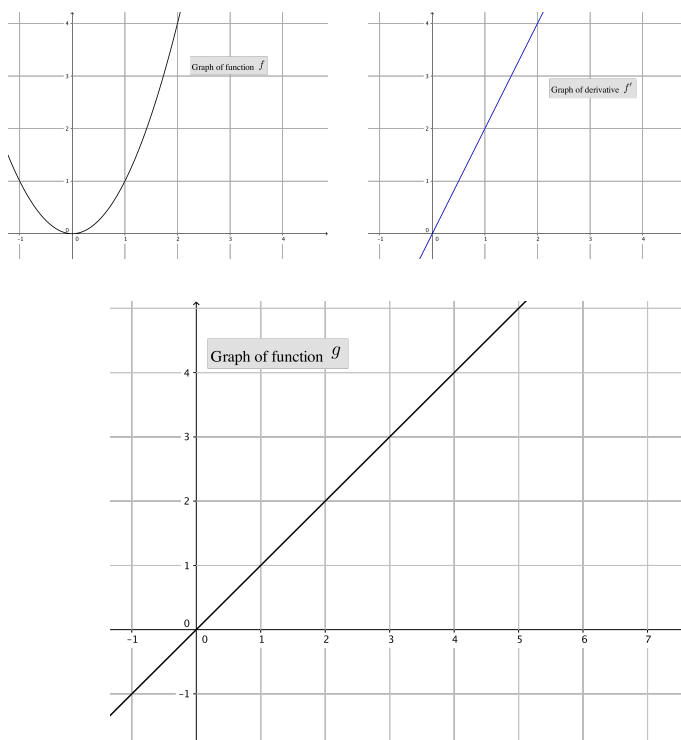


Problem 4

Problem 4

Problem 1 Use the given graphs of f and g and their accompanying derivatives to answer the following questions.

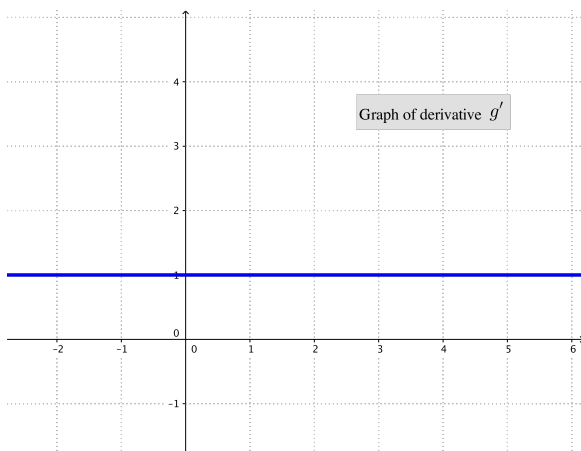


- (a) Write an equation for the tangent line to f at $x = 2$.

Explanation. $f(2) = 4$, $f'(2) = 4$, $y - 4 = 4(x - 2)$, $y = 4x - 4$

- (b) Draw the graph of g'

Problem 4



Explanation.

- (c) Compute the value of the $(5f + 3g)'(2)$.

Explanation.

$$\begin{aligned}(5f + 3g)'(2) &= 5 \cdot f'(2) + 3 \cdot g'(2) \\ &= 5 \cdot 4 + 3 \cdot 1 \\ &= 20 + 3 = 23.\end{aligned}$$

- (d) Find the equation of the tangent line to the graph of $(5f + 3g)$ at the point where $x = 2$.

Explanation.

$$\begin{aligned}(5f + 3g)(2) &= 5 \cdot f(2) + 3 \cdot g(2) \\ &= 5 \cdot 4 + 3 \cdot 2 \\ &= 20 + 6 = 26\end{aligned}$$

$$\begin{aligned}y - 26 &= 23(x - 2) \\ y &= 23x - 20\end{aligned}$$

- (e) Use the given graph of f' and g' to find the following:

- (i) A formula for f'

Explanation. $f'(x) = 2x$. The graph shows a linear function through the point $(0, 0)$ and $(1, 2)$.

- (ii) A formula for g'

Explanation. $g'(x) = 1$. The graph shows a linear function with slope 0 and through the point $(0, 1)$.

- (f) Find the expression for $(5f + 3g)'(x)$.

Explanation.

$$\begin{aligned}(5f + 3g)'(x) &= 5 \cdot f'(x) + 3 \cdot g'(x) \\ &= 5 \cdot 2x + 3 \cdot 1 \\ &= 10x + 3\end{aligned}$$